

# Data bases June 2020 exam

1)

Consider the following schema:

SALES(product, quantity, year)

Here we want to see how many people bought staff from 2001 to 2010. What's the problem with this query?

```
SELECT Year, SUM(Quantity)
```

```
FROM Sales
```

```
WHERE Year >= 2001 AND Year <=2010
```

```
GROUP BY Year
```

(Domanda aperta)

Answer: if there is no record for 2005, then the tuple (2005, 0) won't appear.

2)

Consider the following Database:

S(A, B)

| A | B  |
|---|----|
| 1 | 10 |
| 2 | 20 |
| 3 | 30 |

T(B, C, D)

| B | C    | D    |
|---|------|------|
| 1 | 10   | NULL |
| 3 | NULL | w    |

U (D, E)

| D    | E    |
|------|------|
| NULL | f    |
| f    | w    |
| m    | NULL |

And the following query:

```
SELECT S.B+10, T.C+10, T.D
```

```
FROM S
```

```
JOIN T ON S.B*10 >= T.C*10
```

Write the result of this query.

(Domanda aperta)

Answer:

11, 20, NULL

12, 20, NULL

13, 20, NULL

3-4-5 schema)

Consider the following schema:

Suppliers(sid:integer, sname: string, address: string)

Parts(pid: integer, pname: string, color:string)

Catalog(sid:integer, pid: integer, cost: real)

3)

Explain in natural language (less than 25 words the query) the following query:

```
SELECT C.sid
FROM Catalog C
WHERE NOT EXISTS (
    SELECT P.pid
    FROM Parts P
    WHERE NOT EXISTS (
        SELECT C1.sid
        FROM Catalog C1
        WHERE C1.sid = C.sid AND C1.pid = C.pid
    )
)
```

(Domanda aperta)

Answer: This is a division

4-5 were the same question)

Explain in natural language (less than 25 words the query) the following query:

```
SELECT C.pid
FROM Catalog C
MINUS
SELECT C1.pid
FROM Catalog C1, Catalog C2
WHERE C1.cost < C2.cost
```

(Domanda aperta)

Answer: only the most expensive remains. It's the same of MAX(...)

6)

Consider the relation R(A,B,C,D,E,F) with the following functional dependencies:

```
AB -> C
BC -> AD
D -> E
DF -> B
```

What is the DA+?

Answer: DAE

Wrong answers:

- ABCDEF
- DA
- ABCD

7)

Consider the relation R(A,B,C,D,F). It is decomposed into the relations R1(A,B,C) and R2(A,D,F). The attribute A is the key for R2. Which one of the following statements is true?

Answer: The decomposition is lossless-join because A is the key for R2

Wrong answers:

- The decomposition is not lossless-join
- The decomposition is lossless-join because there are no functional dependencies in R
- The decomposition is BCNF but not lossless join

8)

Consider the relation R(A,B,C,D,F) that has no functional dependencies. Which one of the following statements is true?

Answer: R is not in BCNF because there are no functional dependencies

Wrong answers:

- R is in BCNF because it has no key
- R is not in BCNF because it has not key
- Cannot say anything about whether it is in BCNF or not

9) Consider the relation R(W,X,Y,Z). What is the best index to use in order to make run as fast as possible the query?

```
SELECT R.X
FROM R
WHERE R.W > 10
```

Answer: Better use no index at all

Wrong answers:

- Hash index on (X,Y,Z)
- Hash index on X
- B+tree on (X,Y,W)
- B+tree on X

10) Consider the following two tables:

Students(Sid, Name, Login, Age, GPA)

Enrolled(Sid, Cid, Grade)

Sid is the student id and Cid is the course id. Consider the following query, which one of the following statements is correct about it?

```
SELECT DISTINCT Cid
FROM Enrolled
WHERE Grade = 'C'
```

Answer: Returns the ids of the courses that have at least one student that received a grade C in it

Wrong answers:

- None of the other options
- Returns the ids of the courses for every time a student that received a grade C in it

11) Consider the two relations  $R(A, B)$  and  $S(A, B)$  and the relational algebra query  $R \bowtie S$ . Which one of the following statements is true?

Answer: The query produces the intersection of the two relations

Wrong answers:

- The query produces the union of the two relations
- The query produces the full outer join of the two relations
- None of the other options is true

12)

Consider the following two queries:

Query 1 `SELECT * FROM Employee WHERE Salary <> NULL`

Query 2 `SELECT * FROM Employee WHERE Salary > NULL`

Scegli un'alternativa:

- The 2 queries give different results and the union of their results is not always the EMPLOYEE relation
- The two queries give exactly the same result
- The results of the two queries partially overlap
- Cannot say anything because the results depends on the data
- The 2 queries give different results and the union of their results is the EMPLOYEE relation

Answer: The two queries give exactly the same result

13)

Consider the two relations  $R$  and  $S$  and the following relational algebra query:  $R - (R - S)$ . To which of the following queries is it equivalent

Scegli un'alternativa:

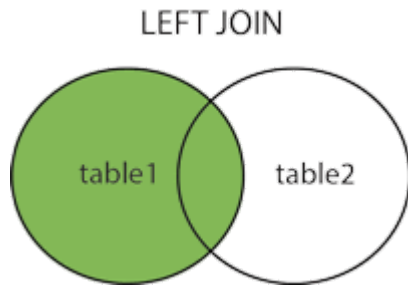
- $R - S$
- None of the above
- $R \cap S$
- $S$
- $R \cup S$

Answer:  $R \cap S$

14)

Consider two relations (tables)  $A$  and  $B$  that have exactly the same schema. The attribute  $C$  is the key in each of the two tables. A circle in the figure below represents the tuples that belong to a table, and each point in the circle (I guess was a typo) represents a tuple in that

table. For example, a point inside the circle B represents a tuple that is in the table B. Select the query amount those below that returns the tuples in the green part.



Scegli un'alternativa:

- $A - (A \cup B)$
- $B - A$
- $A - (A \bowtie B)$
- None of the above
- $A - (A \cap B)$

Answer:  $A - (A \bowtie B)$

15)

Consider the following schedule:

$r1(A) : w1(A) : r2(A) : w2(B) : r3(C) : w3(A)$

Which of the following statements is correct regarding its conflict serializability?

Scegli un'alternativa:

- It is not serializable
- It is serializable to T3 T1 T2
- It is serializable to T1 T2 T3
- It is serializable to T1 T2 T2

Answer: It is serializable to T1 T2 T3

16)

Consider the following schedule:

$r1(A) : w1(A) : r2(A) : w1(B) : r3(C) : w3(A)$

Which of the following statements is correct regarding its conflict serializability?

Scegli un'alternativa:

- It is serializable to T1 T2 T3
- It is not serializable
- It is serializable to T1 T3 T2
- It is serializable to T3 T1 T2

Answer: It is serializable to T1 T2 T3

17)

Consider the following schedule:

$r_1(A) : w_1(B) : r_2(B) : w_1(B) : r_3(C) : w_3(A)$

Which of the following statements is correct regarding its conflict serializability?

Scegli un'alternativa:

- It is serializable to T1 T3 T2
- It is serializable to T1 T2 T3
- It is not serializable
- It is serializable to T3 T1 T2

Answer: It is not serializable

18)

The "SELECT" operation in SQL and the "PROJECT" operation in relational algebra are equivalent (meaning that they produce always the same results if applied on the same relation). The specific statement is:

Scegli un'alternativa:

- True
- False

Answer: False